

FAKE NEWS DETECTION USING MACHINE LEARNING

AI/ML MINI PROJECT REPORT

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2. Project Overview

Problem Statement

The rapid growth of online news platforms and social media has increased the spread of fake news and misinformation. Fake news can influence public opinion, create confusion, and cause social and political issues. Manually verifying every news article is time-consuming and inefficient. Therefore, an automated system is required to identify whether a news article is real or fake using machine learning techniques. This project aims to classify news articles accurately and help users verify information quickly.

Objective

The main objective of this project is to develop a machine learning-based Fake News Detection System that can automatically classify news articles as Real News or Fake News. The system uses Natural Language Processing (NLP) and machine learning algorithms to analyze the content of news articles and provide accurate predictions.

Technologies Used

- Python
- Pandas
- NumPy
- Scikit-learn
- TF-IDF Vectorizer
- Passive Aggressive Classifier
- Streamlit
- Machine Learning
- Natural Language Processing (NLP)

3. Dataset Details

Dataset Name: Fake and Real News Dataset

Dataset Source: Kaggle Dataset

The dataset was downloaded from Kaggle and contains a collection of real and fake news articles used for training and testing the machine learning model.

Number of Records

- Fake News Articles: **23,481**
- Real News Articles: **21,417**
- Total Records: **44,898**

Dataset Files

1. **Fake.csv** – Contains fake news articles.
2. **True.csv** – Contains real news articles.

Features (Columns)

- **Title** – Headline of the news article.
- **Text** – Main content of the news article.
- **Subject** – Category of the news article.
- **Date** – Publication date of the article.

Dataset Link

<https://www.kaggle.com/datasets/clmentbisailon/fake-and-real-news-dataset>

4. Project Workflow

The Fake News Detection System follows a series of steps to classify news articles as real or fake using machine learning techniques.

Step 1: Data Collection

The Fake News Dataset was downloaded from Kaggle. The dataset consists of two files: **Fake.csv** containing fake news articles and **True.csv** containing real news articles.

Step 2: Data Loading

The datasets were loaded into Python using the Pandas library for further processing and analysis.

Step 3: Data Labeling

Labels were assigned to the datasets:

- Fake News → 0
- Real News → 1

This helps the machine learning model distinguish between fake and real news articles.

Step 4: Data Merging and Shuffling

Both datasets were combined into a single dataset and shuffled randomly to avoid bias during training.

Step 5: Feature Extraction

The title and text of each news article were combined and converted into numerical features using the **TF-IDF (Term Frequency-Inverse Document Frequency) Vectorizer**.

Step 6: Data Splitting

The dataset was divided into:

- Training Data (80%)
- Testing Data (20%)

The training data was used to train the model, while the testing data was used to evaluate its performance.

Step 7: Model Training

The **Passive Aggressive Classifier** algorithm was trained using the TF-IDF transformed news articles to learn patterns associated with fake and real news.

Step 8: Model Testing

The trained model was tested on unseen data to measure its prediction capability and overall performance.

Step 9: Accuracy Evaluation

The model's accuracy was calculated using the Accuracy Score metric. The project achieved an accuracy of approximately **99%** on the testing dataset.

Step 10: News Prediction

Users can enter any news article text through the Streamlit interface, and the trained model predicts whether the news is **Real** or **Fake**.

Step 11: User Interface Deployment

A simple and interactive web application was developed using Streamlit to allow users to test news articles easily.

Dataset Collection



Data Loading



Data Labeling



Data Preprocessing



TF-IDF Vectorization



Train-Test Split



Model Training



Model Testing



Accuracy Evaluation



News Prediction



Streamlit User Interface

5. Model / Algorithm Used

1. TF-IDF (Term Frequency–Inverse Document Frequency) Vectorizer

The TF-IDF Vectorizer is a feature extraction technique used to convert textual data into numerical form so that it can be processed by machine learning algorithms. It measures the importance of words in a document relative to the entire dataset.

Why TF-IDF Was Selected

- Converts text data into machine-readable numerical vectors.
- Reduces the impact of commonly occurring words such as "the", "is", and "and".
- Highlights important words that help distinguish between fake and real news.
- Efficient and widely used in Natural Language Processing (NLP) applications.

2. Passive Aggressive Classifier

The Passive Aggressive Classifier is a supervised machine learning algorithm used for classification tasks. It updates the model only when it makes incorrect predictions, making it highly efficient for text classification problems.

Why Passive Aggressive Classifier Was Selected

- Provides high accuracy for text classification tasks.
- Works efficiently with large datasets.
- Fast training and prediction speed.
- Suitable for real-time applications.
- Performs well with TF-IDF feature vectors.

Working of the Model

1. The news dataset is loaded and labeled.
2. News titles and article text are combined.
3. TF-IDF Vectorizer converts text into numerical vectors.
4. The dataset is split into training and testing sets.

5. The Passive Aggressive Classifier is trained using the training data.
6. The trained model predicts whether a news article is Real or Fake.
7. Model performance is evaluated using accuracy score.

Model Performance

The trained model achieved an accuracy of approximately **99%** on the testing dataset, demonstrating its effectiveness in identifying fake and real news articles.

6. Implementation

Code Implementation

The Fake News Detection System was implemented using Python and Machine Learning libraries. The project loads the dataset, preprocesses the news articles, converts text into numerical features using TF-IDF Vectorization, trains a Passive Aggressive Classifier, and predicts whether a news article is real or fake.

Important Code Snippet

```
vectorizer = TfidfVectorizer(stop_words='english', max_df=0.7)

model = PassiveAggressiveClassifier(max_iter=1000)

model.fit(tfidf_train, y_train)

prediction = model.predict(input_data)
```


Screenshots:

	A	B	C	D	E	F	G	H	I	J	K
1	title	text	subject	date							
2	Donald Trump Sends Out Embarrassing New Year's Eve Mess	Donald Trump just couldn't wish all Americans a Happy New Year and instead	News	December 31, 2017							
3	Drunk Bragging Trump Staffer Started Russian Collusion Investigation	House Intelligence Committee Chairman Devin Nunes is going to have	News	December 31, 2017							
4	Sheriff David Clarke Becomes An Internet Joke For Threatening To Kill	On Friday, it was revealed that former Milwaukee Sheriff David Clarke,	News	December 30, 2017							
5	Trump Is So Obsessed He Even Has Obama's Name Coded In On Christmas Day	Donald Trump announced that he would be back to	News	December 29, 2017							
6	Pope Francis Just Called Out Donald Trump During His Christmas	Pope Francis used his annual Christmas Day message to rebuke Donald	News	December 28, 2017							
7	Racist Alabama Cops Brutalize Black Boy While He Is In Handcuff	The number of cases of cops brutalizing and killing people of color seer	News	December 25, 2017							
8	Fresh Off The Golf Course, Trump Lashes Out At FBI Deputy Director	Donald Trump spent a good portion of his day at his golf club, marking it	News	December 23, 2017							
9	Trump Said Some INSANELY Racist Stuff Inside The Oval Office	In the wake of yet another court decision that derailed Donald Trump's	News	December 23, 2017							
10	Former CIA Director Slams Trump Over UN Bullying	Openly Suggests Many people have raised the alarm regarding the fact that Donald Trump	News	December 22, 2017							
11	WATCH: Brand-New Pro-Trump Ad Features So Much A**	Kissin' Just when you might have thought we'd get a break from watching poop	News	December 21, 2017							
12	Papa John's Founder Retires	Figures Out Racism is Bad For A Centerpiece of Donald Trump's campaign, and now his presidency	h News	December 21, 2017							
13	WATCH: Paul Ryan Just Told Us He Doesn't Care About Strong Republicans	are working overtime trying to sell their scam of a tax bill to	News	December 21, 2017							
14	Bad News For Trump & Mitch McConnell Says No To Repealing	Republicans have had seven years to come up with a viable replacement	News	December 21, 2017							
15	WATCH: Lindsey Graham Trashes Media For Portraying Trump As The	media has been talking all day about Trump and the Republican Pa	News	December 20, 2017							
16	Heiress To Disney Empire Knows GOP Scammed Us & SHREDs	Abigail Disney is an heiress with brass ovaries who will profit from the C	News	December 20, 2017							
17	Tone Deaf Trump: Congrats Rep. Scalise On Losing Weight After	Donald Trump just signed the GOP tax scam into law. Of course, that m	News	December 20, 2017							
18	The Internet Brutally Mocks Disney's New Trump Robot At Walt Disney	World News	December 19, 2017								
19	Mueller Spokesman Just F---ed Up Donald Trump's Christmas	Trump supporters and the so-called president's favorite network are la	News	December 17, 2017							
20	SNL Hilariously Mocks Accused Child Molester Roy Moore For Lo	Right now, the whole world is looking at the shocking fact that Democr	News	December 17, 2017							
21	Republican Senator Gets Dragged For Going After Robert Mueller	Senate Majority Whip John Cornyn (R-TX) thought it would be a good ide	News	December 16, 2017							
22	In A Heartless Rebuke To Victims, Trump Invites NRA To Xmas Pa	rt almost seems like Donald Trump is trolling America at this point. In th	News	December 16, 2017							
23	KY GOP State Rep. Commits Suicide Over Allegations He Molested	In this #METOO moment, many powerful men are being toppled. It spar	News	December 13, 2017							
24	Meghan McCain Tweets The Most AMAZING Response To Doug J	As a Democrat won a Senate seat in deep-red Alabama, social media c	News	December 12, 2017							
25	CNN CALLS IT: A Democrat Will Represent Alabama In The Senat	Alabama is a notoriously deep red state. It's a place where Democrats'	News	December 12, 2017							
26	White House: It Wasn't Sexist For Trump To Slut-Shame Sen. I	A backlash ensued after Donald Trump launched a sexist rant against K	News	December 12, 2017							
27	Despicable Trump Suggests Female Senator Would & Do Anyth	Donald Trump is afraid of strong, powerful women. He is a horrific miss	News	December 12, 2017							

	A	B	C	D	E	F	G	H
1	title	text	subject	date				
2	As U.S. budget fight looms, Republicans flip their fiscal script	WASHINGTON (Reuters) - The head of a conservative Republ	politicsNews	December 31, 2017				
3	U.S. military to accept transgender recruits on Monday: Pentagon	WASHINGTON (Reuters) - Transgender people will be allowe	politicsNews	December 29, 2017				
4	Senior U.S. Republican senator: 'Let Mr. Mueller do his job'	WASHINGTON (Reuters) - The special counsel investigation o	politicsNews	December 31, 2017				
5	FBI Russia probe helped by Australian diplomat tip-off: NYT	WASHINGTON (Reuters) - Trump campaign adviser George P.	politicsNews	December 30, 2017				
6	Trump wants Postal Service to charge 'much more' for Amazon shipments	SEATTLE-WASHINGTON (Reuters) - President Donald Trump	politicsNews	December 29, 2017				
7	White House, Congress prepare for talks on spending, immigration	WEST PALM BEACH, Fla./WASHINGTON (Reuters) - The White	politicsNews	December 29, 2017				
8	Trump says Russia probe will be fair, but timeline unclear: NYT	WEST PALM BEACH, Fla. (Reuters) - President Donald Trump	politicsNews	December 29, 2017				
9	Factbox: Trump on Twitter (Dec 29) - Approval rating, Amazon	The following statements were posted to the verified Twitter	politicsNews	December 29, 2017				
10	Trump on Twitter (Dec 28) - Global Warming	The following statements were posted to the verified Twitter	politicsNews	December 29, 2017				
11	Alabama official to certify Senator-elect Jones today despite challenge: CNN	WASHINGTON (Reuters) - Alabama Secretary of State John M	politicsNews	December 28, 2017				
12	Jones certified U.S. Senate winner despite Moore challenge	(Reuters) - Alabama officials on Thursday certified Democrat	politicsNews	December 28, 2017				
13	New York governor questions the constitutionality of federal tax overhaul	NEW YORK/WASHINGTON (Reuters) - The new U.S. tax code	politicsNews	December 28, 2017				
14	Factbox: Trump on Twitter (Dec 28) - Vanity Fair, Hillary Clinton	The following statements were posted to the verified Twitter	politicsNews	December 28, 2017				
15	Trump on Twitter (Dec 27) - Trump, Iraq, Syria	The following statements were posted to the verified Twitter	politicsNews	December 28, 2017				
16	Man says he delivered manure to Munchin to protest new U.S. tax law	(In Dec. 25 story, in second paragraph, corrects name of Stro	politicsNews	December 25, 2017				
17	Virginia officials postpone lottery drawing to decide tied statehouse election	(Reuters) - A lottery drawing to settle a tied Virginia legislat	politicsNews	December 27, 2017				
18	U.S. lawmakers question businessman at 2016 Trump Tower meeting: sources	WASHINGTON (Reuters) - A Georgian-American businessman	politicsNews	December 27, 2017				
19	Trump on Twitter (Dec 26) - Hillary Clinton, Tax Cut Bill	The following statements were posted to the verified Twitter	politicsNews	December 26, 2017				
20	U.S. appeals court rejects challenge to Trump voter fraud panel	(Reuters) - A U.S. appeals court in Washington on Tuesday upl	politicsNews	December 26, 2017				
21	Treasury Secretary Mnuchin was sent gift-wrapped box of horse manure: reports	(Reuters) - A gift-wrapped package addressed to U.S. Treasur	politicsNews	December 24, 2017				
22	Federal judge partially lifts Trump's latest refugee restrictions	WASHINGTON (Reuters) - A federal judge in Seattle partially b	politicsNews	December 24, 2017				
23	Exclusive: U.S. memo weakens guidelines for protecting immigrant children in court	NEW YORK (Reuters) - The U.S. Justice Department has issue	politicsNews	December 23, 2017				
24	Trump travel ban should not apply to people with strong U.S. ties: court	(Reuters) - A U.S. appeals court on Friday said President Don	politicsNews	December 23, 2017				
25	Second court rejects Trump bid to stop transgender military recruits	WASHINGTON (Reuters) - A federal appeals court in Washing	politicsNews	December 23, 2017				
26	Failed vote to oust president shakes up Peru's politics	LIMA (Reuters) - Peru's President Pedro Pablo Kuczynski c	politicsNews	December 23, 2017				
27	Trump signs tax, government spending bills into law	WASHINGTON (Reuters) - U.S. President Donald Trump signe	politicsNews	December 22, 2017				

```
File Edit Selection View Go Run Terminal Help FakeNewsDetector

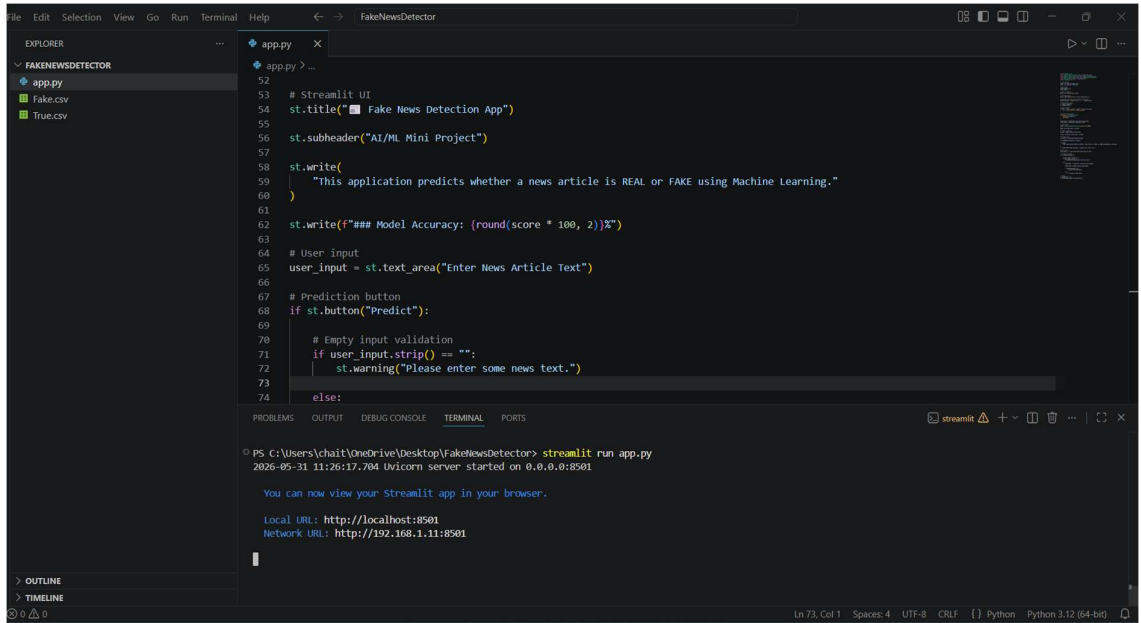
EXPLORER
FAKE_NEWS_DETECTOR
  app.py
  fake.csv
  true.csv

app.py
1 import pandas as pd
2 from sklearn.model_selection import train_test_split
3 from sklearn.feature_extraction.text import TfidfVectorizer
4 from sklearn.linear_model import PassiveAggressiveClassifier
5 from sklearn.metrics import accuracy_score
6 import streamlit as st
7
8 # Load Datasets
9 fake = pd.read_csv("fake.csv")
10 true = pd.read_csv("true.csv")
11
12 # Add labels
13 fake["label"] = 0
14 true["label"] = 1
15
16 # Combine datasets
17 data = pd.concat([fake, true])
18
19 # Shuffle dataset
20 data = data.sample(frac=1).reset_index(drop=True)
21
22 # Combine title and text for better prediction
23 data["content"] = data["title"] + " " + data["text"]
24
25 # Features and labels
26 x = data["content"]
27 y = data["label"]
28
29 # Split dataset
30 x_train, x_test, y_train, y_test = train_test_split(
31     x, y, test_size=0.2, random_state=42
32 )
33
34 # TF-IDF Vectorization
35 vectorizer = TfidfVectorizer(
36     stop_words="english",
37     max_df=0.7
```

```
File Edit Selection View Go Run Terminal Help FakeNewsDetector

EXPLORER
FAKE_NEWS_DETECTOR
  app.py
  fake.csv
  true.csv

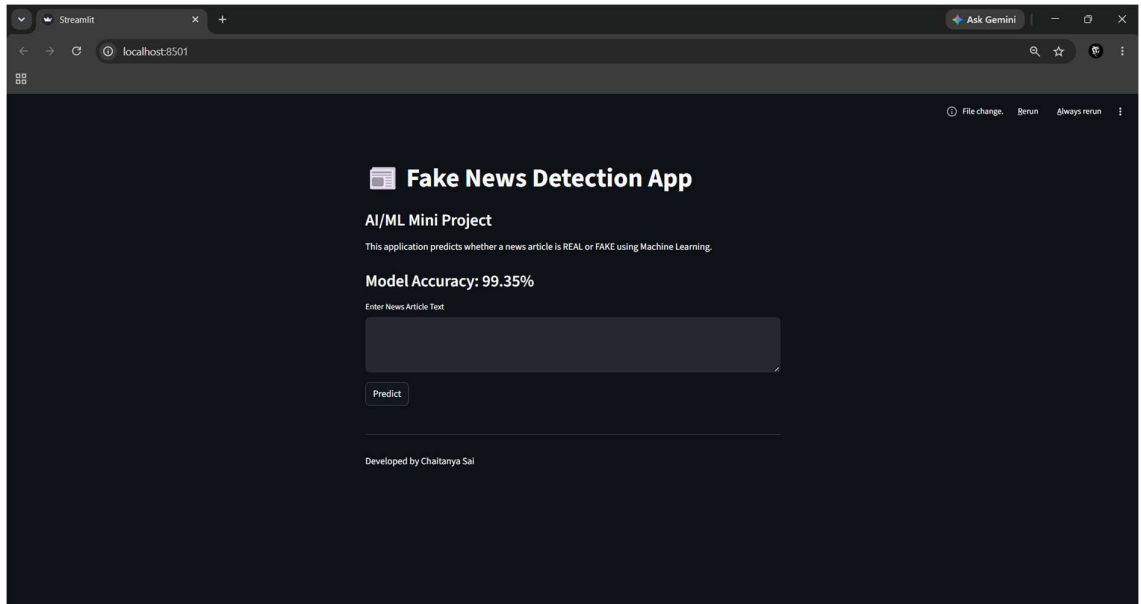
app.py
52
53 # Streamlit UI
54 st.title("Fake News Detection App")
55
56 st.subheader("AI/ML Mini Project")
57
58 st.write(
59     "This application predicts whether a news article is REAL or FAKE using Machine Learning."
60 )
61
62 st.write(f"### Model Accuracy: {round(score * 100, 2)}%")
63
64 # User input
65 user_input = st.text_area("Enter News Article Text")
66
67 # Prediction button
68 if st.button("Predict"):
69
70     # Empty input validation
71     if user_input.strip() == "":
72         st.warning("Please enter some news text.")
73
74     else:
75         input_data = vectorizer.transform([user_input])
76
77         prediction = model.predict(input_data)
78
79         if prediction[0] == 0:
80             st.error("🚫 Fake News")
81
82         else:
83             st.success("✅ Real News")
84
85 # Footer
86 st.markdown("----")
87 st.write("Developed by Chaitanya Sai")
```

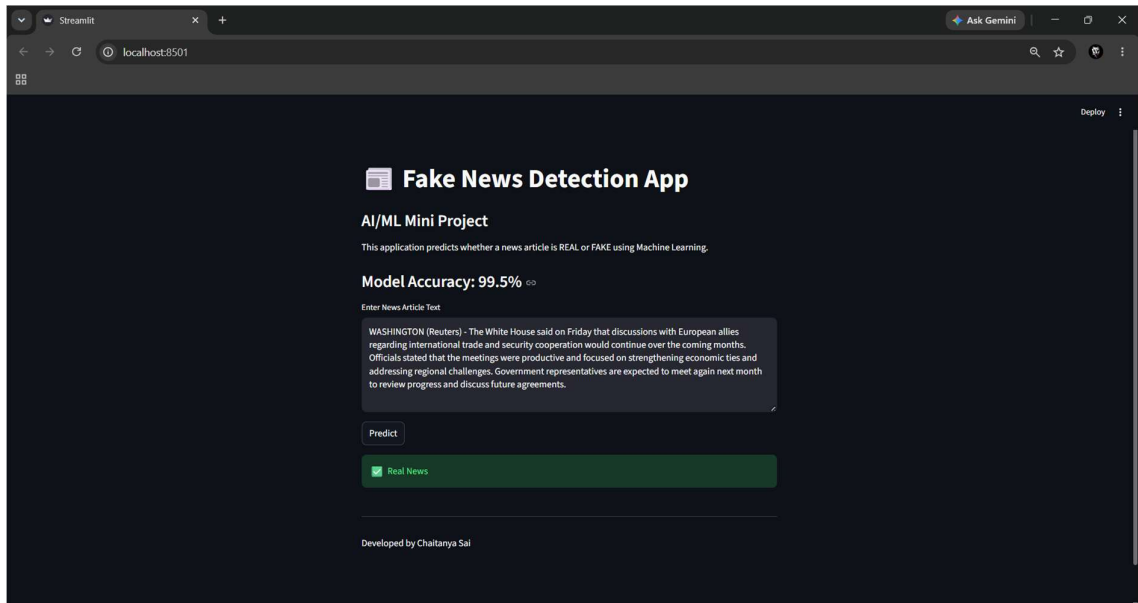
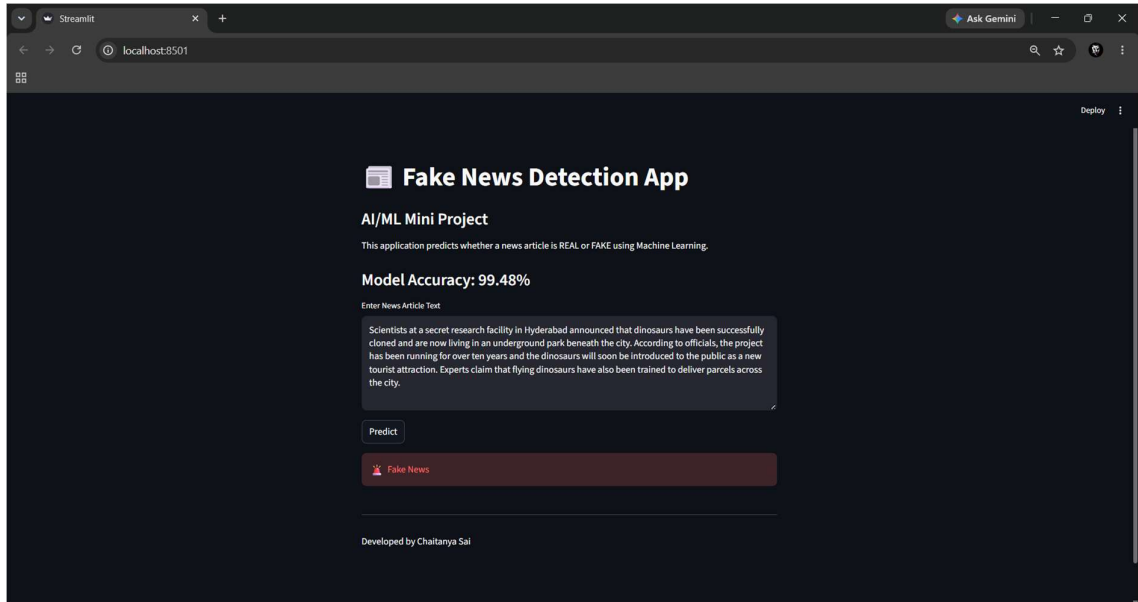


The image shows a Visual Studio Code editor window with a file explorer on the left showing a project named 'FAKE NEWS DETECTOR' containing 'app.py', 'fake.csv', and 'True.csv'. The main editor displays the 'app.py' file with the following Python code:

```
52
53 # Streamlit UI
54 st.title("Fake News Detection App")
55
56 st.subheader("AI/ML Mini Project")
57
58 st.write(
59     "This application predicts whether a news article is REAL or FAKE using Machine Learning."
60 )
61
62 st.write(f"### Model Accuracy: {round(score * 100, 2)}%")
63
64 # User input
65 user_input = st.text_area("Enter News Article Text")
66
67 # Prediction button
68 if st.button("Predict"):
69
70     # Empty input validation
71     if user_input.strip() == "":
72         st.warning("Please enter some news text.")
73
74     else:
```

The terminal at the bottom shows the command 'streamlit run app.py' being executed, with output indicating the server started on 0.0.0.0:8501 and providing local and network URLs.





Training Process

1. Dataset Loading
2. Data Labelling
3. Data Merging and Shuffling
4. TF-IDF Feature Extraction
5. Train-Test Split
6. Model Training
7. Model Evaluation
8. News Prediction

Accuracy / Results

The model was trained using the Passive Aggressive Classifier and achieved an accuracy of approximately **99.4%** on the testing dataset. The application successfully predicts whether a news article is real or fake through a Streamlit-based user interface.

7. Results

Accuracy

- Training and Testing Accuracy: **99.4%**

Output Predictions

The system accepts a news article as input and predicts one of the following:

- Real News
- Fake News

Sample Outputs

Input:

"Scientists confirmed that dinosaurs are secretly living under Hyderabad metro stations."

Output:

Fake News

Input:

"The Reserve Bank increased interest rates by 0.25 percent to control inflation."

Output:

Real News

Graphs or Charts

The following visualizations can be included:

- Dataset Distribution Chart (Real vs Fake News)
- Accuracy Performance Chart
- Prediction Result Screenshots

Performance Analysis

The model demonstrated excellent performance on the testing dataset. TF-IDF effectively extracted meaningful textual features, while the Passive Aggressive Classifier provided fast and accurate classification. The system successfully identified patterns associated with fake and real news articles and achieved high prediction accuracy.

8. Challenges Faced

1. Understanding the structure of the dataset and preparing it for training.
2. Combining and labeling fake and real news records correctly.
3. Converting textual news data into numerical features using TF-IDF.
4. Improving prediction quality for unseen news articles.
5. Developing and integrating the Streamlit user interface.
6. Managing large text datasets efficiently during model training.
7. Testing the model with different types of news articles and validating results.

9. Conclusion

The Fake News Detection System was successfully developed using Machine Learning and Natural Language Processing techniques. The system can classify news articles as real or fake based on their textual content. By using TF-IDF Vectorization and the Passive Aggressive Classifier, the model achieved an accuracy of approximately 99.4%.

This project demonstrates the practical application of machine learning in combating misinformation and improving the reliability of digital news consumption.

Future Improvements

- Use advanced deep learning models such as BERT or RoBERTa.
- Train the model on more recent datasets.
- Add support for news article URLs.
- Deploy the application on a cloud platform.
- Improve prediction accuracy for unseen real-world news articles.

10. References

Dataset

Fake and Real News Dataset (Kaggle)

<https://www.kaggle.com/clmentbisailon/fake-and-real-news-dataset>

Documentation: Scikit-learn Documentation-<https://scikit-learn.org>

Streamlit Documentation-<https://streamlit.io>

Pandas Documentation-<https://pandas.pydata.org>

Learning Resources: Machine Learning Tutorials

GeeksforGeeks<https://www.geeksforgeeks.org>

Machine Learning with Python – W3Schools: <https://www.w3schools.com>

Official Python Documentation: <https://docs.python.org>